The CIs are a second approach to check for multicollinearity (Belsley et al., 2005). A CI higher than 30 signals moderate to strong multicollinearity, while a CI<10 indicates weak multicollinearity. When two or more variables both have a very high variance decomposition proportion (>0.9) for the same CI, they are strongly correlated with each other. Within our data, the CIs indicated no problematic multicollinearity issues (Table 11).

Table 11: **Condition Indices (CIs), calculated for the continuous variables used in the landslide susceptibility model**

	Condition index	Slope	Elevation	Profile curvature	Planar curvature	North exposure	East exposure	PGA	Distance to rivers	Distance to faults	Road buffer	>35% Clay
1	**1.000**	0.007	0.002	0.000	0.000	0.000	0.000	0.004	0.010	0.012	0.006	0.013
2	**1.817**	0.000	0.000	0.319	0.311	0.000	0.000	0.000	0.000	0.000	0.004	0.000
3	**2.110**	0.000	0.000	0.000	0.000	0.521	0.452	0.000	0.000	0.000	0.013	0.000
4	**2.127**	0.000	0.000	0.000	0.000	0.465	0.539	0.000	0.000	0.000	0.000	0.000
5	**2.254**	0.004	0.000	0.007	0.000	0.011	0.007	0.001	0.001	0.000	0.927	0.003
6	**2.412**	0.006	0.000	0.001	0.000	0.000	0.001	0.000	0.779	0.012	0.000	0.022
7	**2.664**	0.000	0.000	0.670	0.687	0.001	0.000	0.000	0.002	0.000	0.003	0.000
8	**3.332**	0.075	0.005	0.001	0.001	0.001	0.000	0.033	0.023	0.312	0.012	0.216
9	**3.870**	0.004	0.002	0.000	0.000	0.000	0.001	0.001	0.003	0.459	0.011	0.728
10	**6.203**	0.755	0.020	0.001	0.000	0.000	0.000	0.301	0.084	0.001	0.025	0.016
11	**11.324**	0.149	**0.971**	0.000	0.000	0.000	0.000	0.661	0.097	0.204	0.000	0.001

No problematic multicollinearity was observed.
Note that the morphological parameters were derived here from the 10 m resolution DEM.

Reference

Belsley, D. A., Kuh, E., Welsch, R. E., 2005. Regression Diagnostics. Hoboken, New Jersey: John Wiley & Sons. doi:10.1002/0471725153